

2023-Paper

m -dependent Potential

$$V_m^-(x) = V(x) - 2 \left[\frac{\mathcal{H}_m''}{\mathcal{H}_m} - \left(\frac{\mathcal{H}_m'}{\mathcal{H}_m} \right)^2 + 1 \right], \quad (1)$$

$$\psi_{0,m}^-(x) = \left(\frac{2^m m!}{\sqrt{\pi}} \right)^{\frac{1}{2}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)}, \quad (2)$$

$$\psi_{n+1,m}^-(x) = \frac{1}{\sqrt{E_{n,m}^+ 2^n n! \sqrt{\pi}}} \frac{e^{-\frac{x^2}{2}}}{\mathcal{H}_m(x)} y_{n+1}^m(x), \quad n = 0, 1, 2, \dots, \quad (3)$$

$$E_{n+1,m}^- = 2(n + m + 1), \quad n = 0, 1, 2, \dots, \quad \text{and } E_{0,m}^- = 0. \quad (4)$$

m and λ -dependent Potential

$$\hat{V}_m^-(\lambda, x) = V_m^-(x) - 2 \frac{d^2}{dx^2} \ln [\mathcal{I}_m(x) + \lambda], \quad (5)$$

$$\mathcal{I}_m(x) = \left(\frac{2^m m!}{\sqrt{\pi}} \right) \int_{-\infty}^x \left[\frac{e^{-\frac{x'^2}{2}}}{\mathcal{H}_m(x')} \right]^2 dx'. \quad (6)$$

$$\hat{\psi}_{0,m}^-(\lambda, x) = \sqrt{\lambda(1 + \lambda)} \frac{\psi_{0,m}^-(x)}{\mathcal{I}_m(x) + \lambda}. \quad (7)$$

$$\begin{aligned} \hat{\psi}_{n+1,m}^-(\lambda, x) &= \frac{1}{E_{n+1,m}^-} \hat{A}^\dagger A \psi_{n+1,m}^-(x), \quad n = 0, 1, 2, \dots, \\ &= \frac{e^{-\frac{x^2}{2}}}{\sqrt{2(n + m + 1) 2^n n! \sqrt{\pi}}} \left(\frac{y_{n+1}^m(x)}{\mathcal{H}_m(x)} + H_n(x) \frac{d}{dx} \ln [\mathcal{I}_m(x) + \lambda] \right) \end{aligned} \quad (8)$$

When $\lambda = 0$ or 1 (P or AM) the spectrum is strictly isospectral to the spectrum of starting potential.

$$E_{n,m}^+ = 2(n + m + 1), \quad n = 0, 1, 2, \dots \quad (9)$$

2024-Paper

$$V^+(x) = \frac{1}{4}\omega_x^2 x^2, \quad -\infty < x < \infty, \quad (10)$$

$$\psi_n^+(x) \propto e^{-\frac{\omega_x}{4}x^2} H_n\left(\sqrt{\frac{\omega_x}{2}}x\right)$$

$$E_n^+ = \left(n + \frac{1}{2}\right)\omega_x, \quad n = 0, 1, 2, \dots \quad (11)$$

$$V^-(x, m) = V^+(x) - 2 \left[\frac{\mathcal{H}_m''\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} - \left[\frac{\mathcal{H}_m'\left(\sqrt{\frac{\omega_x}{2}}x\right)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} \right]^2 + \frac{\omega_x}{2} \right], \quad -\infty < x < \infty \quad (12)$$

$$\psi_0^-(x, m) = \phi_m^{-1}(x) \propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)}, \quad m = 0, 2, 4, \dots \quad (13)$$

$$\text{and } \psi_{n+1}^-(x, m) \propto \frac{e^{-\frac{\omega_x}{4}x^2}}{\mathcal{H}_m\left(\sqrt{\frac{\omega_x}{2}}x\right)} \hat{H}_{n+1,m}\left(\sqrt{\frac{\omega_x}{2}}x\right), \quad n = 0, 1, 2, \dots$$

$$\hat{H}_{n+1,m}(z) = \left[\mathcal{H}_m(z) H_{n+1}(z) + H_n(z) \frac{d}{dz} \mathcal{H}_m(z) \right], \quad (14)$$

$$E_{n+1,m}^- = E_{n,m}^+ = E_n^+ - \epsilon_m = (n + m + 1)\omega_x \quad \text{with} \quad E_{0,m}^- = 0. \quad (15)$$

$$V_h^+(x) = \begin{cases} \frac{1}{4}\omega_x^2 x^2 & , \quad x > 0 \\ \infty & , \quad x \leq 0. \end{cases} \quad (16)$$

$$\psi_{h,n}^+(x, \alpha) \propto x^{\alpha+\frac{1}{2}} e^{-\frac{\omega_x}{4}x^2} L_n^{(\alpha)}\left(\frac{\omega_x}{2}x^2\right); \quad n = 0, 1, 2, \dots \quad (17)$$

$$E_{h,n}^+(\alpha) = (2n + 1 + \alpha)\omega_x \quad \text{and} \quad \epsilon_{h,m}(\alpha) = -(2m + 1 + \alpha)\omega_x \quad (18)$$

$$E_{h,n,m}^+(\alpha) = E_{h,n}^+(\alpha) - \epsilon_{h,m}(\alpha) = 2(n + m + \alpha + 1)\omega_x. \quad (19)$$

$$V_h^-(x, m, \frac{1}{2}) = V_h^+(x) - 2 \left(\frac{\left[x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right) \right]''}{x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} - \left(\frac{\left[x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right) \right]'}{x L_m^{(\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} \right)^2 + \frac{\omega_x}{2} \right); \quad m = 0, 1, 2, \dots \quad (20)$$

$$V_h^-(x, m, -\frac{1}{2}) = V_h^+(x) - 2 \left(\frac{L_m^{(-\frac{1}{2})''} \left(-\frac{\omega_x}{2} x^2 \right)}{L_m^{(-\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} - \left(\frac{L_m^{(-\frac{1}{2})'} \left(-\frac{\omega_x}{2} x^2 \right)}{L_m^{(-\frac{1}{2})} \left(-\frac{\omega_x}{2} x^2 \right)} \right)^2 + \frac{\omega_x}{2} \right); \quad m = 0, 1, 2, \dots \quad (21)$$

$$\psi_{h,n}^-(x, m, \alpha) \propto x^{\alpha+\frac{3}{2}} e^{-\frac{\omega_x}{4}x^2} \frac{\hat{L}_{n,m}^{(\alpha)}(\frac{\omega_x}{2}x^2)}{L_m^{(\alpha)}(-\frac{\omega_x}{2}x^2)} \quad (22)$$

$$\hat{L}_{n,m}^{(\alpha)}(s) = L_m^{(\alpha)}(-s)L_n^{(\alpha+1)}(s) + L_{m-1}^{(\alpha+1)}(-s)L_n^{(\alpha)}(s). \quad (23)$$

$$E_{h,n,m}^-(\alpha) = E_{h,n,m}^+(\alpha) = 2(n+m+1+\alpha)\omega_x. \quad (24)$$

In 2D

$$V^+(x, y) = \frac{1}{4}\omega_x^2x^2 + \frac{1}{4}\omega_y^2y^2; \quad -\infty < x < \infty, -\infty < y < \infty \quad (25)$$

$$\begin{aligned} \psi_{n_1,n_2}^+(x, y) &\propto \psi_{n_1}^+(x)\psi_{n_2}^+(y) \\ &\propto e^{-\frac{\omega_x x^2 + \omega_y y^2}{4}} H_{n_1}\left(\sqrt{\frac{\omega_x}{2}}x\right) H_{n_2}\left(\sqrt{\frac{\omega_y}{2}}y\right) \quad n_1, n_2 = 0, 1, 2, \dots, \end{aligned} \quad (26)$$

$$E_{n_1,n_2}^+ = E_{n_1}^+ + E_{n_2}^+ \quad (27)$$

$$V^-(x, y, m_1, m_2) = V^-(x, m_1) + V^-(y, m_2) \quad (28)$$

$$\begin{aligned} \psi_{0,0}^-(x, y, m_1, m_2) &= \phi_{m_1}^{-1}(x)\phi_{m_2}^{-1}(y), \\ \text{and } \psi_{n_1+1,n_2+1}^-(x, y, m_1, m_2) &= \psi_{n_1+1}^-(x, m_1)\psi_{n_2+1}^-(y, m_2), \quad n_1, n_2 = 0, 1, 2, \dots \end{aligned} \quad (29)$$

$$E_{n_1+1,n_2+1,m_1,m_2}^- = [(n_1+m_1+1)\omega_x + (n_2+m_2+1)\omega_y] \quad \text{and} \quad E_{0,0,m_1,m_2}^- = 0. \quad (30)$$

$$\psi_{h,n_1,n_2}^+(x, y, \alpha, \beta) \propto x^{\alpha+\frac{1}{2}}y^{\beta+\frac{1}{2}}e^{-\frac{\omega_x x^2 + \omega_y y^2}{4}} L_{n_1}^{(\alpha)}\left(\frac{\omega_x}{2}x^2\right) L_{n_2}^{(\beta)}\left(\frac{\omega_y}{2}y^2\right); \quad n_1, n_2 = 0, 1, 2, \dots \quad (31)$$

$$E_{h,n_1,n_2,\alpha,\beta}^+ = [(2n_1+\alpha+1)\omega_x + (2n_2+\beta+1)\omega_y] - \epsilon_{h,m_1,m_2}, \quad (32)$$

$$V_h^-(x, y, m_1, m_2, \frac{1}{2}) = V_h^-(x, m_1, \frac{1}{2}) + V_h^-(y, m_2, \frac{1}{2}) \quad (33)$$

$$\psi_{h,n_1,n_2}^-(x, y, m_1, m_2, \frac{1}{2}) \propto \psi_{n_1}^-(x, m_1, \frac{1}{2})\psi_{n_2}^-(y, m_2, \frac{1}{2}) \quad (34)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1, m_2, \frac{1}{2}) = [(2n_1+2m_1+3)\omega_x + (2n_2+2m_2+3)\omega_y] \quad (35)$$

$$V_h^-(x, y, m_1, m_2, -\frac{1}{2}) = V_h^-(x, m_1, -\frac{1}{2}) + V_h^-(y, m_2, -\frac{1}{2}) \quad (36)$$

$$\psi_{h,n_1,n_2}^-(x, y, m_1, m_2, -\frac{1}{2}) \propto \psi_{h,n_1}^-(x, m_1, -\frac{1}{2}) \psi_{h,n_2}^-(y, m_2, -\frac{1}{2}) \quad (37)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1, m_2, \frac{1}{2}) = [(2n_1 + 2m_1 + 1)\omega_x + (2n_2 + 2m_2 + 1)\omega_y] . \quad (38)$$

$$V_h^-(x, y, m_1, m_2, \frac{1}{2}, -\frac{1}{2}) = V_h^-(x, m_1, \frac{1}{2}) + V_h^-(y, m_2, -\frac{1}{2}) \quad (39)$$

$$\psi_{h,n_1,n_2}^-(x, y, m_1, m_2, \frac{1}{2}, -\frac{1}{2}) \propto \psi_{h,n_1}^-(x, m_1, \frac{1}{2}) \psi_{h,n_2}^-(y, m_2, -\frac{1}{2}) \quad (40)$$

$$\text{and } E_{h,n_1,n_2}^-(m_1, m_2, \frac{1}{2}, -\frac{1}{2}) = [(2n_1 + 2m_1 + 3)\omega_x + (2n_2 + 2m_2 + 1)\omega_y] \quad (41)$$

$$V_{fh}^-(x, y, m_1, m_2, \beta) = V^-(x, m_1) + V_h^-(y, m_2, \beta), \quad (42)$$

$$\psi_{fh,0,0}^-(x, y, m_1, m_2, \beta) \propto \psi_0^-(x, m_1) \psi_{h,0}^-(y, m_2, \beta) \quad (43)$$

$$\psi_{fh,n_1+1,n_2}^-(x, y, m_1, m_2, \beta) \propto \psi_{n_1+1}^-(x, m_1) \psi_{h,n_2}^-(y, m_2, \beta); \quad (44)$$

$$E_{fh,0,0}^-(m_2, \beta) = 2(m_2 + \beta + 1)\omega_y. \quad (45)$$

$$\begin{aligned} E_{fh,n_1+1,n_2}^-(m_1, m_2, \beta) &= E_{n_1}^+(m_1) + E_{h,n_2}^+(m_2, \beta) \\ &= (n_1 + m_1 + 1)\omega_x + 2(n_2 + m_2 + \beta + 1)\omega_y \end{aligned} \quad (46)$$

$$V_{fh}^-(x, y, m_1, m_2, 1/2) = V^-(x, m_1) + V_h^-(y, m_2, \frac{1}{2}) \quad (47)$$

$$\psi_{fh,0,0}^-(m_1, m_2, \frac{1}{2}) \propto \psi_0^-(x, m_1) \psi_{h,0}^-(y, m_2, \frac{1}{2}) \quad (48)$$

$$\psi_{n_1+1,n_2}^-(m_1, m_2, \frac{1}{2}) \propto \psi_{n_1+1}^-(x, m_1) \psi_{h,n_2}^-(y, m_2, \frac{1}{2}) \quad (49)$$

$$E_{0,0}^-(m_1, m_2, \frac{1}{2}) = (2m_2 + 3)\omega_y \quad (50)$$

$$E_{n_1+1,n_2}^-(m_1, m_2, \frac{1}{2}) = (n_1 + 2m_1 + 1)\omega_x + (2n_2 + 2m_2 + 3)\omega_y \quad (51)$$

$$V_{fh}^-(x, y, m_1, m_2, -1/2) = V^-(x, m_1) + V_h^-(y, m_2, -\frac{1}{2}) \quad (52)$$

$$\psi_{fh,0,0}^-(m_1, m_2, -\frac{1}{2}) \propto \psi_0^-(x, m_1) \psi_{h,0}^-(y, m_2, -\frac{1}{2}) \quad (53)$$

$$\psi_{n_1+1,n_2}^-(m_1, m_2, -\frac{1}{2}) \propto \psi_{n_1+1}^-(x, m_1) \psi_{h,n_2}^-(y, m_2, -\frac{1}{2}) \quad (54)$$

$$E_{0,0}^-(m_1, m_2, -\frac{1}{2}) = (2m_2 + 1) \omega_y \quad (55)$$

$$E_{n_1+1,n_2}^-(m_1, m_2, -\frac{1}{2}) = (n_1 + 2m_1 + 1) \omega_x + (2n_2 + 2m_2 + 1) \omega_y \quad (56)$$

In 3D

$$V^+(x, y, z) = \frac{1}{4} \omega^2 (x^2 + y^2) + \frac{1}{4} \omega_z^2 z^2; \quad -\infty < x, y, z < \infty \quad (57)$$

$$V^+(r, z) = \frac{1}{4} \omega^2 r^2 + \frac{1}{4} \omega_z^2 z^2 \quad (58)$$

$$\left[- \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \right) + V^+(r, z) \right] \psi^+(r, \phi, z) = E \psi^+(r, \phi, z), \quad (59)$$

$$\psi^+(r, \phi, z) = \frac{e^{i\gamma\phi} \zeta(r, z)}{r^{\frac{1}{2}}}. \quad (60)$$

$$\zeta(r, z) = R(r) Z(z). \quad (61)$$

$$\left[- \frac{\partial^2}{\partial r^2} - \frac{\partial^2}{\partial z^2} + V_{eff}^+(r, z) \right] \zeta(r, z) = E \zeta(r, z), \quad (62)$$

$$V_{eff}^+(r, z) = V^+(r, z) + \frac{\gamma^2 - 1/4}{r^2} \quad \text{with} \quad \gamma = 0, \pm 1, \pm 2, \dots \quad (63)$$

$$R(r) \rightarrow R_{n_1}(r) \propto r^{|\gamma| + \frac{1}{2}} e^{-\frac{\omega}{4} r^2} L_{n_1}^{(|\gamma|)} \left(\frac{1}{2} \omega r^2 \right); n_1 = 0, 1, 2, \dots \quad (64)$$

$$Z(z) \rightarrow Z_{n_2}(z) \propto e^{-\frac{\omega_z}{4} z^2} H_{n_2} \left(\sqrt{\frac{\omega_z}{2}} z \right); n_2 = 0, 1, 2, \dots \quad (65)$$

$$E \rightarrow E_{n_1, n_2} = (2n_1 + |\gamma| + 1) \omega + \left(n_2 + \frac{1}{2} \right) \omega_z. \quad (66)$$

$$E_{n_1, n_2}^+ = E_{n_1, n_2} - \epsilon_{m_1, m_2} = 2(n_1 + m_1 + |\gamma| + 1) \omega + (n_2 + m_2 + 1) \omega_z. \quad (67)$$

$$\begin{aligned}
V_{eff}^-(m_1, m_2, r, z) &= \frac{1}{4}\omega^2 r^2 + \frac{\gamma^2 - \frac{1}{4}}{r^2} + \frac{2|\gamma|}{r^2} - (2m+1)\omega + \frac{1}{r^2} \\
&+ \frac{2r^2\omega^2 L_{m_1-1}^{(|\gamma|+1)}\left(-\frac{\omega r^2}{2}\right)^2}{L_{m_1}^{(|\gamma|)}\left(-\frac{\omega r^2}{2}\right)^2} \\
&+ \frac{\omega\left((2|\gamma| + r^2\omega)L_{m_1-1}^{(|\gamma|+1)}\left(-\frac{\omega r^2}{2}\right) - r^2\omega L_{m-2}^{(|\gamma|+2)}\left(-\frac{\omega r^2}{2}\right)\right)}{L_{m_1}^{(|\gamma|)}\left(-\frac{\omega r^2}{2}\right)} \\
&+ \frac{1}{2}\omega_z^2 z^2 - 2\left[\frac{\mathcal{H}_{m_2}''\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} - \left[\frac{\mathcal{H}_{m_2}'\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}\right]^2 + \frac{\omega_z}{2}\right] \\
\zeta_{n_1, n_2+1}^-(r, z, m_1, m_2) &\propto r^{|\gamma|+\frac{3}{2}} e^{-\frac{\omega r^2}{4}} \frac{\hat{L}_{n_1, m_1}^{(\gamma)}\left(\frac{1}{2}\omega r^2\right)}{L_{m_1}^{(\gamma)}\left(-\frac{1}{2}\omega r^2\right)} e^{-\frac{\omega_z z^2}{2}} \frac{\hat{H}_{n_z+1, m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} \\
E_{n_1, n_2+1}^-(m_1, m_2) &= 2(n_1 + m_1 + |\gamma| + 1)\omega + (n_2 + m_2 + 1)\omega_z
\end{aligned} \tag{68}$$

$$\begin{aligned}
\zeta_{0,0}^-(r, z, m_1, m_2) &\propto r^{|\gamma|+\frac{3}{2}} \frac{e^{-\frac{\omega r^2}{4}} L_{m_1}^{(|\gamma|+1)}\left(-\frac{1}{2}\omega r^2\right)}{L_{m_1}^{(|\gamma|)}\left(-\frac{1}{2}\omega r^2\right)} \times \frac{e^{-\frac{\omega_z z^2}{4}}}{\mathcal{H}_{m_2}\left(\sqrt{\frac{\omega_z}{2}}z\right)} \\
\text{and } E_{0,0,m_1,m_2}^- &= 2(m_1 + |\gamma| + 1)\omega
\end{aligned} \tag{69}$$

2025-Paper

$$V^l(x) = V(x) + \lambda_0 x, \quad V(x) = \frac{1}{4}\omega_1^2 \quad (70)$$

$$\tilde{V}^l(\tilde{x}) = \frac{1}{4}\omega_1^2 \tilde{x}^2 - \frac{\lambda_0^2}{\omega_1^2} \quad \text{where} \quad \tilde{x} = x + \frac{2\lambda_0}{\omega_1^2}, \quad (71)$$

$$E \rightarrow \tilde{E}_n^l(\omega_1) = \left(n + \frac{1}{2}\right) \omega_1 + \frac{\gamma_0^2}{\omega_1^2}; \quad n = 0, 1, 2, \dots \quad (72)$$

$$\tilde{\psi}_n^l(\tilde{x}) = \psi_n(\tilde{x}), \quad \text{Here,} \quad \tilde{\omega}_1 = \omega_1. \quad (73)$$

1D

$$V_{RE,m}(x) = V(x) + V_{rat,m}(x), \quad -\infty < x < \infty \quad (74)$$

$$V_{rat,m}(x) = -2 \left[\frac{\mathcal{H}_m''(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} - \left[\frac{\mathcal{H}_m'(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \right]^2 + \frac{\omega_1}{2} \right] \quad (75)$$

$$\begin{aligned} \psi_{RE,m,0}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \\ \text{and } \psi_{RE,m,n+1}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \hat{H}_{m,n+1} \left(\sqrt{\frac{\omega_1}{2}}x \right), \end{aligned} \quad (76)$$

$$\hat{H}_{m,n+1}(x) = \mathcal{H}_m(x)H_{n+1}(x) + H_n(x)\frac{d}{dx}\mathcal{H}_m(x), \quad (77)$$

$$E_{RE,m,n+1}(\omega_1) = (n + m + 1)\omega_1 \quad \text{with} \quad E_{RE,m,0}(\omega_1) = 0. \quad (78)$$

$$\tilde{V}_{RE,m}^l(\tilde{x}) = \tilde{V}^l(\tilde{x}) + V_{rat,m}(\tilde{x}), \quad -\infty < x < \infty \quad (79)$$

$$\begin{aligned} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}\tilde{x})} \\ \text{and } \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}\tilde{x})} \hat{H}_{m,n+1} \left(\sqrt{\frac{\omega_1}{2}}\tilde{x} \right). \end{aligned} \quad (80)$$

$$\tilde{E}_{RE,m,n+1}^l(\omega_1) = (n + m + 1)\omega_1 \quad \text{with} \quad \tilde{E}_{RE,m,0}^l(\omega_1) = 0. \quad (81)$$

2D

$$V(x, y) = \frac{1}{4}\omega_1^2 x^2 + \frac{1}{4}\omega_2^2 y^2 + \frac{\lambda}{2}xy. \quad (82)$$

$$\begin{aligned} x &= a\tilde{x} + b\tilde{y}, \\ y &= -b\tilde{x} + a\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = ax - by, \\ \tilde{y} = bx + ay, \end{cases} \quad (83)$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 - \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \\ \text{and } \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 + \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \end{aligned} \quad (84)$$

$$\tilde{V}(\tilde{x}, \tilde{y}) = \tilde{V}(\tilde{x}) + \tilde{V}(\tilde{y}). \quad (85)$$

$$\tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{n_1}(\tilde{x}) \tilde{\psi}_{n_2}(\tilde{y}), \quad n_1, n_2 = 0, 1, 2, \dots, \quad (86)$$

$$\text{and } \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) = \left(n_1 + \frac{1}{2}\right) \tilde{\omega}_1 + \left(n_2 + \frac{1}{2}\right) \tilde{\omega}_2, \quad (87)$$

$$\tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y}) = \tilde{V}_{RE, m_1}(\tilde{x}) + \tilde{V}_{RE, m_2}(\tilde{y}), \quad (88)$$

$$\tilde{\psi}_{RE, m_1, m_2, 0, 0}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{RE, m_1, 0}(\tilde{x}) \tilde{\psi}_{RE, m_2, 0}(\tilde{y}) \quad (89)$$

$$\tilde{\psi}_{RE, m_1, m_2, n_1+1, n_2+1}(x, y) \propto \tilde{\psi}_{RE, m_1, n_1+1}(\tilde{x}) \tilde{\psi}_{RE, m_2, n_2+1}(\tilde{y}) \quad (90)$$

$$\text{and } \tilde{E}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) = \tilde{E}_{RE, m_1, n_1+1}(\tilde{\omega}_1) + \tilde{E}_{RE, m_2, n_2+1}(\tilde{\omega}_2) \quad (91)$$

$$\text{with } \tilde{E}_{RE, m_1, m_2, 0, 0}(\tilde{\omega}_1, \tilde{\omega}_2) = 0, \quad n_1, n_2 = 0, 1, 2, \dots,$$

3D-Quadratic+Linear

$$V^{lq}(x, y, z) = \frac{1}{4}(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2) + \lambda_0 z + \frac{\lambda}{2}xy. \quad (92)$$

$$V^{lq}(x, y, z) = V(x, y) + V^l(z), \quad (93)$$

$$\tilde{V}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}(\tilde{x}, \tilde{y}) + \tilde{V}^l(\tilde{z}). \quad (94)$$

$$\tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \tilde{\psi}_{n_3}^l(\tilde{z}) \quad (95)$$

$$\tilde{E}_{n_1, n_2, n_3}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{n_3}(\tilde{\omega}_3) \quad (96)$$

$$\tilde{V}_{RE, m_1, m_2, m_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y}) + \tilde{V}_{RE, m_3}^l(\tilde{z}) \quad (97)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, 0, 0}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, 0}^l(\tilde{z}), \quad (98)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, n_3+1}^l(\tilde{z}) \quad (99)$$

$$\tilde{E}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, n_3+1}^l(\tilde{\omega}_3) \quad (100)$$

$$\tilde{E}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, 0, 0}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, 0}^l(\tilde{\omega}_3)$$

3D-Quadratic Only

$$V^q(x, z, y) = \frac{1}{4} \left(\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx \right). \quad (101)$$

$$V^{q_1}(x, y, z) = \frac{1}{4} \left(\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_2 yz + 2\lambda_3 zx \right) \quad (102)$$

$$\begin{cases} x = ad\tilde{y} + bd\tilde{z} - c\tilde{x}, \\ y = d\tilde{x} + ac\tilde{y} + bc\tilde{z}, \\ z = a\tilde{z} - b\tilde{y}, \end{cases} \begin{cases} \tilde{x} = -cx + d y, \\ \tilde{y} = ad x + acy - bz, \\ \tilde{z} = bd x + bcy + az. \end{cases} \quad (103)$$

$$\tilde{V}^q(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{4} (\tilde{\omega}_1^2 \tilde{x}^2 + \tilde{\omega}_2^2 \tilde{y}^2 + \tilde{\omega}_3^2 \tilde{z}^2) \quad (104)$$

$$\begin{aligned} \tilde{\omega}_1 &= \omega, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 - \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}. \end{aligned} \quad (105)$$

$$\tilde{V}_{RE, m_1, m_2, m_3}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE, m_1}(\tilde{x}) + \tilde{V}_{RE, m_2}(\tilde{y}) + \tilde{V}_{RE, m_3}(\tilde{z}) \quad (106)$$

$$\tilde{\psi}_{RE,m_1,m_2,m_3,0,0,0}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x}) \tilde{\psi}_{RE,m_2,0}(\tilde{y}) \tilde{\psi}_{RE,m_3,0}(\tilde{z})$$

$$\tilde{\psi}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x}) \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y}) \tilde{\psi}_{RE,m_3,n_3+1}(\tilde{z}) \quad (107)$$

$$\begin{aligned} \tilde{E}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) &= \tilde{E}_{RE,m_1,n_1+1}(\tilde{\omega}_1) \\ &+ \tilde{E}_{RE,m_2,n_2+1}(\tilde{\omega}_2) + \tilde{E}_{RE,m_3,n_3+1}(\tilde{\omega}_3) \end{aligned} \quad (108)$$

$$\tilde{E}_{RE,m_1,m_2,m_3,0,0,0}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = 0 \quad (109)$$

$$V^{q_2}(x, y, z) = \frac{1}{4} \left(\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda(yz + zx) \right). \quad (110)$$

$$\begin{cases} x = \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} - \frac{\tilde{x}}{\sqrt{2}}, \\ y = \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} + \frac{\tilde{x}}{\sqrt{2}}, \\ z = a\tilde{z} - b\tilde{y}, \end{cases} \begin{cases} \tilde{x} = \frac{y}{\sqrt{2}} - \frac{x}{\sqrt{2}}, \\ \tilde{y} = \frac{a}{\sqrt{2}}x + \frac{a}{\sqrt{2}}y - bz, \\ \tilde{z} = \frac{b}{\sqrt{2}}x + \frac{b}{\sqrt{2}}y + az. \end{cases} \quad (111)$$

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\omega^2 - \lambda_1}, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 - \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 + \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}. \end{aligned} \quad (112)$$

Uncertainty Calculations: 2023-Paper

$$V_m^-(x) = V(x) - 2 \left[\frac{\mathcal{H}_m''}{\mathcal{H}_m} - \left(\frac{\mathcal{H}_m'}{\mathcal{H}_m} \right)^2 + 1 \right], \quad (113)$$

m	$n = -1$	$n = 0$	$n = 1$
0	0.5	1.5	2.5
2	≈ 0.5172	≈ 1.554	≈ 2.3412
4	≈ 0.5212	≈ 1.6152	≈ 2.2102

Table 1: Uncertainty relation $\Delta x \Delta p$ for the potential $V_m^-(x)$ in the ground state, first excited state, and second excited state for $m = 0, 2, 4$.

$$\hat{V}_m^-(\lambda, x) = V_m^-(x) - 2 \frac{d^2}{dx^2} \ln [\mathcal{I}_m(x) + \lambda], \quad (114)$$

$m/\lambda =$	1×10^{-12}	1×10^{-8}	1×10^{-5}	1×10^{-3}	1×10^{-1}	1×10^2
0	≈ 0.5202	≈ 0.5281	≈ 0.5367	≈ 0.5590	≈ 0.5455	≈ 0.5000
2	≈ 0.5223	≈ 0.5270	≈ 0.5285	≈ 0.5266	≈ 0.5193	≈ 0.5172
4	≈ 0.5228	≈ 0.5252	≈ 0.5246	≈ 0.5232	≈ 0.5215	≈ 0.5212

Table 2: Uncertainty relation $\Delta x \Delta p$ for potential $\hat{V}_m^-(\lambda, x)$ in the ground state for $m = 0, 2, 4$.

Uncertainty Calculations: 2024-Paper

Full Line Oscillator

$$V^+(x) = \frac{1}{4}\omega_x^2 x^2, \quad -\infty < x < \infty, \quad (115)$$

$n \backslash m$	0	2	4	6
0	0.5	0.517247	0.521212	0.522388
1	1.5	1.55415	1.61522	1.66868
2	2.5	2.34124	2.21019	2.12295
3	3.5	3.50248	3.53107	3.56781

Table 3: Uncertainty values for Full Line Oscillator.

Half Line Oscillator

$$V_h^+(x) = \begin{cases} \frac{1}{4}\omega_x^2 x^2 & , \quad x > 0 \\ \infty & , \quad x \leq 0. \end{cases} \quad (116)$$

$$\alpha = -\frac{1}{2}$$

$n \backslash m$	0	1	2	3
0	0.583216	0.68377	0.747385	0.796063
1	1.49105	1.43835	1.44864	1.46897
2	2.37291	2.29944	2.28775	2.29239
3	3.24886	3.17543	3.15381	3.14931

Table 4: Uncertainty values for Half Line Oscillator with $\alpha = -\frac{1}{2}$.

$$\alpha = \frac{1}{2}$$

$n \backslash m$	0	1	2	3
0	0.525237	0.53416	0.539833	0.543865
1	1.46161	1.45048	1.44748	1.44709
2	2.36972	2.34908	2.33979	2.33524
3	3.26505	3.24028	3.22724	3.21972

Table 5: Uncertainty values for Half Line Oscillator with $\alpha = \frac{1}{2}$.